Supplementary materials

An OpenFOAM based solver for three-phase flow in porous media incorporating foam effects

Section 1 presents a detailed derivation of the pressure equation (Eq. 36) discretization process via finite volume method. Next, Section 2 presents validation studies for two and three-phase flow under gravitational effects, based on the works of Durlofsky [9], Abreu [1], and Abreu et al. [2], considering different ratios of gravitational to convective forces. Section 3 provides validation cases addressing capillary pressure effects. It first reproduces the two-phase capillary–gravity equilibrium problem of Horgue et al. [14] and Carrillo et al. [5], solving for the steady-state saturation profile in a one-dimensional water–air column, and subsequently it presents a analysis of a three-phase flow case using the Leverett model for capillary pressure [16] with varying capillary forces, based on the work of Abreu et al. [2]. Section 4 presents a study case based on the laboratory five-spot waterflood experiment of Gaucher and Lindley [11], validating the implementation with respect to source and sink terms under scaled laboratory conditions. The present section focuses on the validation of multiphase foam flow. Together, all the selected validation cases encompass classical reference benchmarks and experimental observations, thereby providing a comprehensive assessment of ImpesFOAM performance under different flow conditions.

1. Finite volume method

In what follows, we present in detail the discretization of the pressure equation, Eq. (36). The discretization of the subsequent governing equations from the PDE system (36)–(40) is carried out analogously. For clarity, we first group some variables as:

$$M_\beta = \mathbb{K}\lambda_\beta = \frac{\mathbb{K}k_{r\beta}}{\mu_\beta}, \quad L_\beta = \mathbb{K}\lambda_\beta\rho_\beta = \frac{\mathbb{K}k_{r\beta}\rho_\beta}{\mu_\beta}, \quad M_t = \sum_\beta M_\beta, \quad \text{and} \quad L_t = \sum_\beta L_\beta. \tag{S1}$$

Using the definitions from Eq. (S1) and integrating the pressure equation, Eq. (36), over a control volume V_C results in:

$$\int_{V_C} \nabla\cdot(-M_t\nabla p_o)\,dV + \int_{V_C}\nabla\cdot(L_t\mathbf{g})\,dV = \int_{V_C}\nabla\cdot\Big[M_g\nabla p_{cgo} - M_w\nabla p_{cow}\Big]dV + \int_{V_C} q_t dV, \tag{S2}$$

and applying the divergence theorem results in:

$$\oint_{\partial V_C} \underbrace{(-M_t\nabla p_o)}_{\text{oil-phase pressure flux}} \cdot d\mathbf{S} + \oint_{\partial V_C}\underbrace{(L_t\mathbf{g})}_{\text{gravity flux}}\cdot d\mathbf{S} = \oint_{\partial V_C}\underbrace{\Big[M_g\nabla p_{cgo} - M_w\nabla p_{cow}\Big]}_{\text{capillary pressure flux}}\cdot d\mathbf{S} + \int_{V_C} q_t dV, \tag{S3}$$

where ∂V_C denotes the surface enclosing the control volume V_C.

Replacing the surface integral over ∂V_C by a summation of the flux terms over each of the ∂V_C faces, when considering ∂V_C to have a finite number of faces, the surface integrals of the fluxes become:

$$\oint_{\partial V_C}(-M_t\nabla p_o)\cdot d\mathbf{S} = \sum_f^{\text{faces }(V_C)}\int_f(-M_t\nabla p_o)\cdot d\mathbf{S} \approx \sum_f^{\text{faces }(V_C)}\underbrace{\sum_{ip}^{N_{ip}(f)}(-M_t\nabla p_o\cdot\mathbf{n})_{ip}\,\omega_{ip}S_f}_{\phi_{p_o}\text{ oil-phase pressure flux at face f}}, \tag{S4}$$

$$\oint_{\partial V_C} (L_t \mathbf{g}) \cdot d\mathbf{S} = \sum_{f}^{\text{faces }(V_C)} \int_f (L_t \mathbf{g}) \cdot d\mathbf{S} \approx \sum_{f}^{\text{faces }(V_C)} \underbrace{\sum_{ip}^{N_{ip}(f)} (L_t \mathbf{g} \cdot \mathbf{n})_{ip} \, \omega_{ip} S_f}_{\phi_g \text{ gravity flux at face f}}, \tag{S5}$$

$$\begin{aligned} \oint_{\partial V_C} \Big[M_g \nabla p_{cgo} - M_w \nabla p_{cow} \Big] \cdot d\mathbf{S} &= \sum_{f}^{\text{faces }(V_C)} \int_f \Big[M_g \nabla p_{cgo} - M_w \nabla p_{cow} \Big] \cdot d\mathbf{S} \approx \\ &\approx \sum_{f}^{\text{faces }(V_C)} \underbrace{\sum_{ip}^{N_{ip}(f)} \left(\Big[M_g \nabla p_{cgo} - M_w \nabla p_{cow} \Big] \cdot \mathbf{n} \right)_{ip} \omega_{ip} S_f}_{\phi_{pc} \text{ capillary pressure flux at face f}}, \end{aligned} \tag{S6}$$

where ip refers to an integration point, $N_{ip}(f)$ is the number of integration points along surface f, S_f is the area of surface f and $\mathbf{n}$ is the unit normal vector outward to face f. To proceed further with the discretization, the surface integral at each face of the volume has to be evaluated. Using Gaussian quadrature, the integral at the face f becomes as shown in Eqs. (S4)–(S6) [10, 17].

The computational cost rises with the number of integration points used in the approximation (see Fig. 1), with accuracy depending on the number of integration points used and the weighing function ω_{ip}. For a simple mid-point integration, only one integration point located at the centroid of the face is used with a weighing function of value equal to 1 (*i.e.*, $ip = \omega_{ip} = 1$) [10, 17], as can be seen in Fig. 1(a). This approximation is second-order accurate and is applicable in two and three dimensions [17], and it is the one used in the scope of this work.

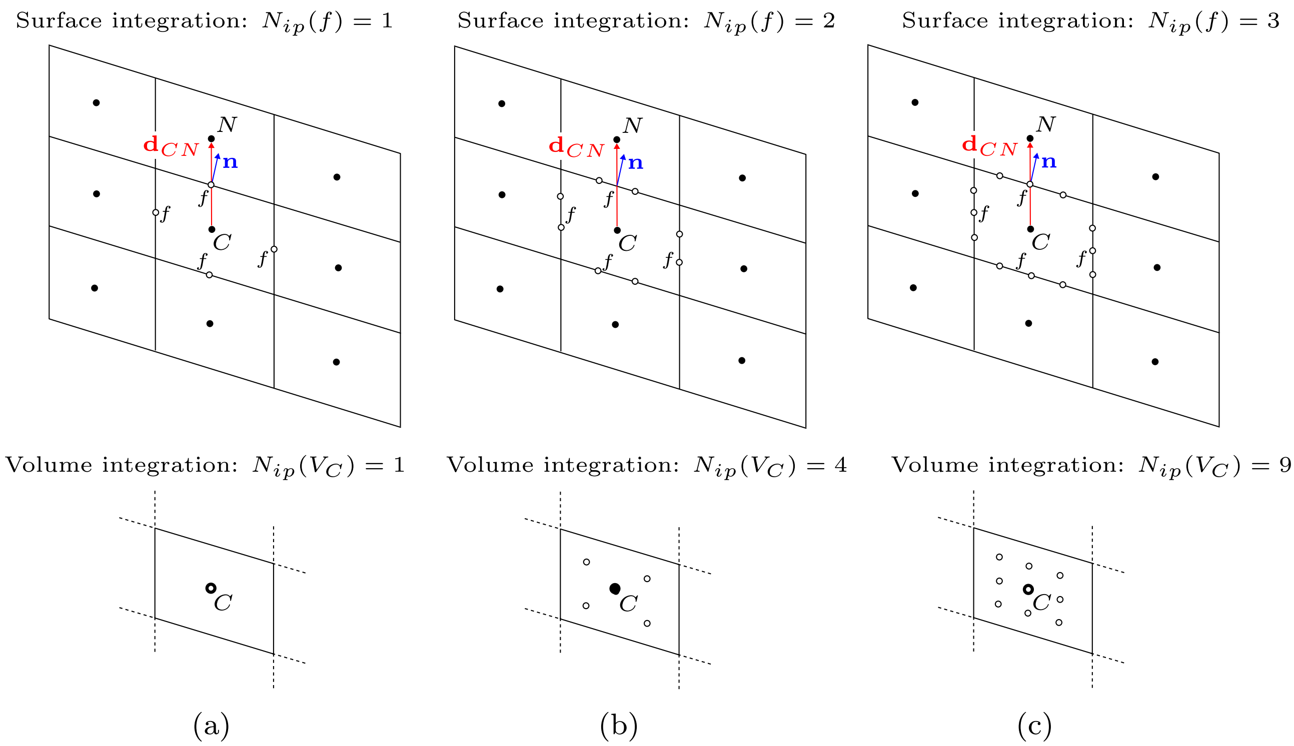


Figure 1: Surface integration of fluxes (upper panels) using (a) one integration point, (b) two integration points, and (c) three integration points. Volume integration of the source terms (lower panels) using (a) one integration point, (b) four integration points, and (c) nine integration points. Adapted from Moukalled et al. [17].

For the source/sink term, $\int_{V_C} q_t dV$, a volume integration is used. Adopting a Gaussian quadrature integration, the volume integral of the source term is computed as:

$$\int_{V_C} q_t dV = \sum_{ip}^{N_{ip}(V_C)} \left(q_{t_{ip}} \omega_{ip} V_C \right). \tag{S7}$$

As with surface flux integration, Fig. 1 shows different options for volume integration with their accuracy depending on the number of integration points used (ip) and the weighing function ω_{ip}. For one point Gaussian quadrature integration, as shown in Fig. 1(a), $ip = \omega_{ip} = 1$ with the integration point located at the centroid of V_C. This approximation is second-order accurate and is applicable in two and three dimensions [17]. The accuracy increases with increasing the number of integration points, but so does the computational cost.

As one can see from Eqs. (S4)–(S6), we need to compute pressure gradients at faces. OpenFOAM reconstructs the face-centered gradient of a scalar field using the geometric relationship between neighboring cells. For a face shared by cells C and N, the face-normal component of the gradient, *e.g.*, $(\nabla p_o \cdot \mathbf{n})_f$, is first approximated by an orthogonal contribution given by the difference between cell-center values, $(p_{o_N} - p_{o_C})/|\mathbf{d}_{CN}|$, which has to be projected onto the face normal $\mathbf{n}$, as shown in Fig. 1. To handle non-orthogonal meshes (*i.e.*, $\mathbf{d}_{CN}$ and $\mathbf{n}$ are not parallel) as the one shown in Fig. 1, a correction term is added based on the interpolated cell-centered gradients, which accounts for the deviation between the line connecting cell centers $\mathbf{d}_{PN}$ and the actual face normal $\mathbf{n}$. This results in a non-orthogonal correction formulation that preserves second-order spatial accuracy for orthogonal grids while maintaining robustness on skewed or unstructured meshes [12].

At this stage, the problem reduces to interpolating the known cell-centered values to the face centers f. This step enables the use of the wide range of numerical interpolation schemes available in OpenFOAM [15, 13], including upwind, linear/limited linear [23], SuperBee [21], harmonic averaging, and van Leer [22], for instance. After the discretization and by using the mid-point integration (one integration point, i.e., $ip = \omega_{ip} = 1$) in the discretized equation, Eq. (36) can be rewritten in its discretized form like:

$$\begin{aligned}\sum_{f}^{\text{faces }(V_C)} \underbrace{\left(-M_t(S_w^n, S_g^n)\nabla p_o^n \cdot \mathbf{n}\right)_f S_f}_{\phi_{p_o} \text{ oil-phase pressure flux at face f}} = \underbrace{q_{t_C} V_C}_{\text{source term}} - \sum_{f}^{\text{faces }(V_C)} \underbrace{\left(L_t(S_w^n, S_g^n)\mathbf{g} \cdot \mathbf{n}\right)_f S_f}_{\phi_g \text{ gravity flux at face f}} - \\ - \sum_{f}^{\text{faces }(V_C)} \underbrace{\left(\left[M_g(S_g^n)\nabla p_{cgo}(S_w^n, S_g^n) - M_w(S_w^n)\nabla p_{cow}(S_w^n, S_g^n)\right] \cdot \mathbf{n}\right)_f S_f}_{\phi_{p_c} \text{ capillary pressure flux at face f}}.\end{aligned} \tag{S8}$$

Applying Eq. (S8) for each control volume V_C in the domain, then a linear algebraic system is assembled and solved implicitly for p_o. In particular, the pressure equation defined in Eq. (36), corresponding to Line 4 of Algorithm 1, is discretized and solved in the form given by Eq. (S8).

Next, we present the spatially discretized form of the water and gas phase saturation equations, while the time discretization scheme was discussed in the previous section, in Eq. (49)–(50). For brevity, we discuss in detail the water saturation equation discretization, while the gas saturation is derived analogously. Integrating $\mathcal{F}_w(p_o^n, S_w^n, S_g^n)$ over a control volume V_C in an element C, we can rewrite the linearization coefficient $\mathcal{F}_w$ as:

$$\int_{V_C} \mathcal{F}_w(p_o^n, S_w^n, S_g^n)\, dV = \int_{V_C} q_w\, dV - \int_{V_C} \nabla \cdot \mathbf{u}_w\, dV. \tag{S9}$$

The source/sink term q_w is integrated over the control volume V_C using one point Gaussian quadrature, as described in Eq. (S7).

By applying the divergence theorem on the integration of the divergence of water velocity $\mathbf{u}_w$, it can be expressed as:

$$\begin{aligned}\int_{V_C} \nabla \cdot \mathbf{u}_w\, dV = \oint_{\partial V_C} \mathbf{u}_w \cdot d\mathbf{S} = \\ = \oint_{\partial V_C} \left[- M_w \nabla p_o + L_w \mathbf{g} + M_w \left(\frac{\partial p_{cow}}{\partial S_w} \nabla S_w + \frac{\partial p_{cow}}{\partial S_g} \nabla S_g \right) \right] \cdot d\mathbf{S}.\end{aligned} \tag{S10}$$

Replacing the surface integral over cell C by a summation of the flux over the faces of ∂V_C, and using the Gaussian quadrature to evaluate the surface integrals at each face f, using the midpoint rule, S_w^{n+1} can be evaluated as:

$$\begin{aligned}\phi\frac{S_w^{n+1}-S_w^n}{\Delta t^{n+1}} &= q_{wC}V_C - \sum_f^{\text{faces }(V_C)}\Bigg[-M_w(S_w^n)\nabla p_o^{n+1} + L_w(S_w^n)\mathbf{g} + \\ &+ M_w(S_w^n)\left(\frac{\partial p_{cow}(S_w^n,S_g^n)}{\partial S_w}\nabla S_w^n + \frac{\partial p_{cow}(S_w^n,S_g^n)}{\partial S_g}\nabla S_g^n\right)\Bigg]_f \cdot \mathbf{n}S_f.\end{aligned} \tag{S11}$$

At this stage, the problem reduces to interpolating the known cell-centered values to the face centers f. Analogously to Eq. (S11), the gas-phase saturation equation is then evaluated in its discretized form as:

$$\begin{aligned}\phi\frac{S_g^{n+1}-S_g^n}{\Delta t^{n}} &= q_{gC}V_C - \sum_f^{\text{faces }(V_C)}\Bigg[-M_g(S_g^n)\nabla p_o^{n+1} + L_g(S_g^n)\mathbf{g} - \\ &- M_g(S_g^n)\left(\frac{\partial p_{cgo}(S_w^n,S_g^n)}{\partial S_w}\nabla S_w^n + \frac{\partial p_{cgo}(S_w^n,S_g^n)}{\partial S_g}\nabla S_g^n\right)\Bigg]_f \cdot \mathbf{n}S_f.\end{aligned} \tag{S12}$$

Next, we present the spatially discretized form of the surfactant transport and dimensionless foam texture equation, while the time discretization has been discussed in the previous section, in Eqs. (55)–(56). For brevity, we focus on the surfactant transport equation, noting that the dimensionless foam texture equation can be discretized in a very similar manner. By integrating $F_{C_s^w}(p_o^n, S_w^n, S_g^n)$ over a control volume V_C, the coefficient $F_{C_s^w}$ can be expressed as:

$$\int_{V_C} F_{C_s^w}(p_o^n, S_w^n, S_g^n)\,dV = \int_{V_C} q_s - \mathcal{F}_w^n C_s^{w\,n}\,dV - \int_{V_C}\nabla\cdot(C_s^{w\,n}\mathbf{u}_w^n)\,dV. \tag{S13}$$

The reaction term $\mathcal{F}_w^n$ can be explicitly evaluated using the available values of S_w^n, S_g^n, and p_o^n from the current time iteration n. Consequently, the volume integral can be computed as:

$$\int_{V_C} q_s - \mathcal{F}_w^n C_s^w\,dV = (q_s - \mathcal{F}_w^n C_s^{w\,n})\,V_C. \tag{S14}$$

Additionally, $\int_{V_C}\nabla\cdot(C_s^{w\,n}\mathbf{u}_w^n)\,dV$ can be approximated analogously to Eq. (S10) and by using the mid rule as follows:

$$\int_{V_C}\nabla\cdot(C_s^w\mathbf{u}_w)\,dV = \oint_{\partial V_C} C_s^w\mathbf{u}_w\cdot d\mathbf{S} = \sum_f^{\text{faces }(V_C)}\int_f C_s^w\mathbf{u}_w\cdot d\mathbf{S} \approx \sum_f^{\text{faces }(V_C)}(C_s^w\,\mathbf{u}_w)\cdot\mathbf{n}S_f, \tag{S15}$$

which leads to the evaluation of $C_s^{w,n+1}$ as:

$$\begin{aligned}H_{C_s^w}(S_w^n)\frac{C_s^{w\,n+1}-C_s^{w\,n}}{\Delta t^{n+1}} &= (q_s^n - \mathcal{F}_w^n C_s^{w\,n})_f\,V_C - \sum_f^{\text{faces }(V_C)} C_{s_f}^{w\,n}\Bigg[-M_w(S_w^n)\nabla p_o^n + \\ &+ L_w(S_w^n)\mathbf{g} + M_w(S_w^n)\left(\frac{\partial p_{cow}(S_w^n,S_g^n)}{\partial S_w}\nabla S_w^n + \frac{\partial p_{cow}(S_w^n,S_g^n)}{\partial S_g}\nabla S_g^n\right)\Bigg]_f\cdot\mathbf{n}S_f.\end{aligned} \tag{S16}$$

At this stage, the problem reduces to interpolating the known cell-centered values to the face centers f. Similarly to that shown in Eq. (S16), the dimensionless foam-texture equation is evaluated in the discretized form as:

$$
\begin{aligned}
H_{n_D}(S_g^n)\frac{n_D^{n+1}-n_D^n}{\Delta t^{n+1}} &= \left(\frac{\phi S_g^n}{n_{max}}[r_g - r_c](S_w^n, n_D^n) + q_f^n - \mathcal{F}_g^n n_D^n\right)_f V_C - \\
&- \sum_f^{\text{faces }(V_C)} n_{D_f}^n \Big[- M_g(S_g^n)\nabla p_o^n + L_g(S_g^n)\mathbf{g} - \\
&- M_g(S_g^n)\left(\frac{\partial p_{cgo}(S_w^n, S_g^n)}{\partial S_w}\nabla S_w^n + \frac{\partial p_{cgo}(S_w^n, S_g^n)}{\partial S_g}\nabla S_g^n\right)\Big]_f \cdot \mathbf{n} S_f.
\end{aligned}
\tag{S17}
$$

2. Benchmark case – Gravitational effects

2.1. Two-phase flow

The validation study in this section follows the work of Durlofsky [9] and de Abreu [8], presenting results for simple one-dimensional water–oil displacement cases ($S_w + S_o = 1$). Firstly, a dimensionless number, G_d, is introduced to quantify the ratio of gravitational to convective effects, defined as:

$$
G_d = \frac{k_c \Delta\rho g}{\mu_o \nu_c}, \tag{S18}
$$

where k_c is the characteristic permeability (e.g., the geometric average of the element permeabilities), ν_c is the characteristic total velocity, $\Delta\rho = \rho_w - \rho_o$, g the gravitational acceleration, and μ_o the dynamic viscosity of the oil.

For the numerical simulations of one-dimensional two-phase flow presented in this section, the following Riemann problem (RP) is considered, where the left and right states are given by:

$$
RP = \begin{cases} S_w^L = 1.0 \quad \text{and} \quad S_w^R = 0, \\ S_o^L = 0 \quad \text{and} \quad S_o^R = 1.0. \end{cases} \tag{S19}
$$

The RP defined in Eq. (S19) represents a domain initially saturated with oil (right state, R) into which only water is injected (left state, L).

The classical quadratic Corey-type model [7] was used as the relative permeability functions, as follows:

$$
k_{rw} = S_w^2 \quad \text{and} \quad k_{ro} = S_o^2 = (1 - S_w)^2. \tag{S20}
$$

For defining the dimensionless number (G_d) in the two tested scenarios, the parameters listed in Table 1 were used. Additionally, the ratio of the dynamic viscosities of the fluids was assumed to be $\mu_o/\mu_w = 5$. The first simulation corresponds to $G_d = 0$ (i.e., no gravity effects, $g = 0$), and the second to $G_d = 2$. The value of μ_o is determined using Eq. (S18) by isolating μ_o, and μ_w is then calculated based on the assumption $\mu_o/\mu_w = 5$.

Parameter	**Value**	**Unit**	**Parameter**	**Value**	**Unit**
ρ_o	0.7	kg/m^3	ρ_w	1.0	kg/m^3
g	9.81	m/s^2	v_c	1.0	m/s
k_c	1.0×10^{-13}	m^2	ϕ	0.25	–
S_{wc}	0	–	S_{ro}	0	–
L	1.0	m	N_x	100	–
Δx	0.01	m	t_{final}	0.125	s
Δt	1×10^{-5}	s			

Table 1: Simulation parameters for the two-phase gravity effects benchmark.

The dimensionless time t_D is evaluated as:

$$t_D = \frac{v_c \cdot t}{(1 - S_{wc} - S_{or})\, L\phi}, \tag{S21}$$

where L is the domain length and ϕ the porosity. Figure 2 shows the our numerical results (ImpesFOAM), which present a good agreement with the literature [9] and with the analytical solution using the fractional flow theory, considering $f_w = \frac{\lambda_w}{\lambda_t} + \frac{K}{q_t}\frac{\lambda_w \lambda_o}{\lambda_t}(\rho_w - \rho_o)g$ [3].

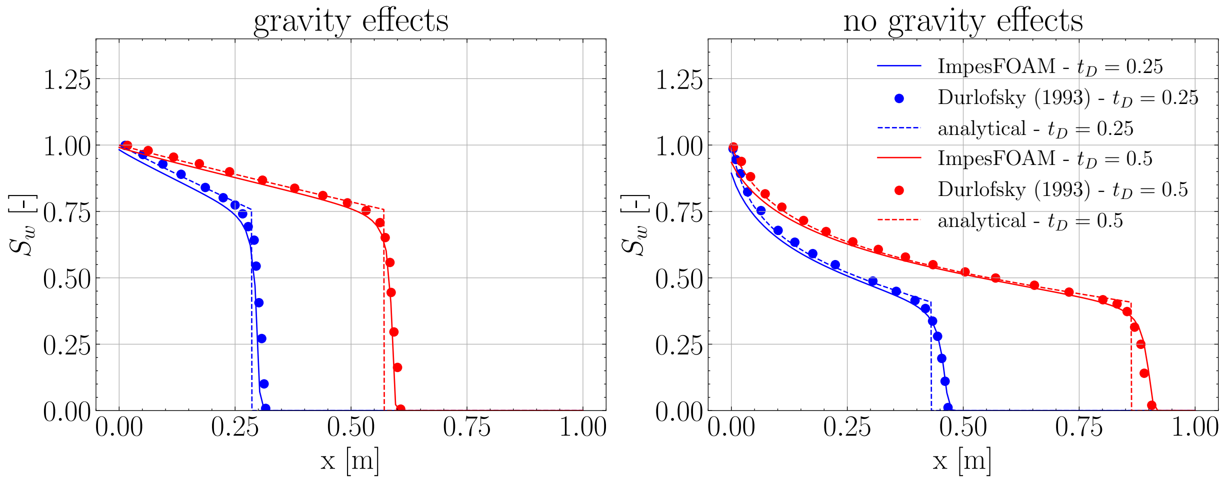


Figure 2: Solution profiles for one-dimensional displacement with $G_d = 2$ (left) and $G_d = 0$ (right) comparing the results from analytical solution [3], numerical results: ImpesFoam (proposed here) and by Durlofsky [9].

Additionally, we compared in Fig. 3 the accuracy of the results when using different interpolation schemes available in OpenFOAM framework. One can observe that the van-Leer [22] method achieved the best results, so it will be considered in subsequent simulations.

2.2. Three-phase flow

For the three-phase flow, we first present numerical simulations of one-dimensional flow using two Riemann problems proposed by Abreu et al. [2] for a three-phase system neglecting gravity and capillary pressure effects. Section 3.2 examine capillary pressure effects on both Riemann problems [2], where the Leverett model [16] is employed for capillary pressure. Conversely, this section examine gravity effects on RP_2 [1]. The Riemam Problems are given by:

$$RP_1 = \begin{cases} S_w^L = 0.613 \quad \text{and} \quad S_w^R = 0.05, \\ S_g^L = 0.387 \quad \text{and} \quad S_g^R = 0.15. \end{cases} \tag{S22}$$

$$RP_2 = \begin{cases} S_w^L = 0.721 \quad \text{and} \quad S_w^R = 0.05, \\ S_g^L = 0.279 \quad \text{and} \quad S_g^R = 0.15. \end{cases} \tag{S23}$$

The classical quadratic Corey-type model [2] was used as the relative permeability functions, as follows:

$$k_{rw} = S_w^2, \quad k_{rg} = S_g^2 \quad \text{and} \quad k_{ro} = S_o^2 = (1 - S_w - Sg)^2. \tag{S24}$$

The parameters employed in the simulations are listed in Table 2. Figure 4 presents the phase saturation profiles at two different time instants, compared with the reference results of Abreu et al. [2]. The results show good agreement with almost indistinguishable profiles for phases saturation.

Next we conduct a case study based on the RP_2 defined in Eq. (S23), now accounting for gravitational effects [1]. The dimensionless group G_d was evaluated using the parameters listed

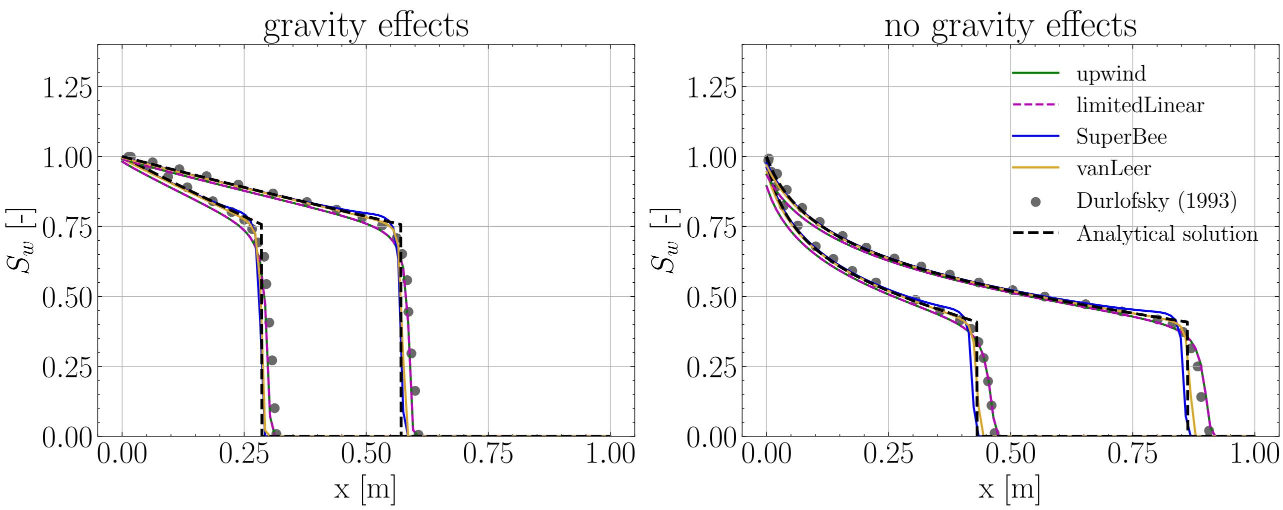


Figure 3: Comparison of the results using different interpolation schemes available in OpenFOAM with those from literature [9] and analytical solution following the methodology outlined by Blunt [3].

Parameter	Value	Unit	Parameter	Value	Unit
ρ_g	5.76×10^{-2}	kg/m^3	ρ_o	0.7	kg/m^3
ρ_w	1.0	kg/m^3	μ_g	0.3	cP
μ_o	1.0	cP	μ_w	0.5	cP
v_c	1.0	m/s	ϕ	1.0	–
$S_{wc} = S_{ro} = S_{gr}$	0	–	$n_w = n_o = n_g$	2.0	–
L_x	0.051	m	N_x	1000	–
Δx	5.1×10^{-5}	m	t_{final}	150	s
Δt	1.0×10^{-4}	s			

Table 2: Simulation parameters for three-phase flow benchmark.

in Table 2. Two scenarios are considered: the first corresponds to $G_d = 0$ (i.e., no gravity effects, $g = 0$), which was already presented in Fig. 4; and the second corresponds to $G_d = 0.0475$, representing a minor influence of gravity. The characteristic permeability k_c is computed from Eq. (S18) as a function of the prescribed G_d value.

Figure 5 presents the saturation profiles for the three phases at a specific time instant, without diffusion, i.e., no capillary pressure, illustrating the agreement between the results obtained from ImpesFOAM solver and those reported by Abreu [1]. It can be seem as a slightly perturbation induced by the dimensionless group $G_d = 0.0475$ (ratio of gravity to convection). The results illustrate the influence of the gravity, which induces a slight perturbation. Both simulations show that the gas saturation front under gravity (blue line) advances slightly ahead compared to the case without gravity (black dashed line). Also, it is observed that the plateau achieved in the gas saturation around $S_g \approx 0.215$ is slightly lower than the one obtained without gravity effects, $S_g \approx 0.219$. The reference case without gravity, $G_d = 0$, is the same profile previously shown in Fig. 4 and is replotted here for comparison. These results are consistent with the expected effect of gravity acting in the opposite direction of the injection direction [1].

3. Benchmark case – Capillary pressure effects

3.1. Two-phase capillary pressure and gravity equilibrium

Now we present a validation case based on the work of Horgue et al. [14] and Carrillo et al. [5] modeling the capillary-gravity equilibrium by solving for the steady state saturation profile of a one-dimensional porous column filled with water and air, as depicted in Fig. 6. Here,

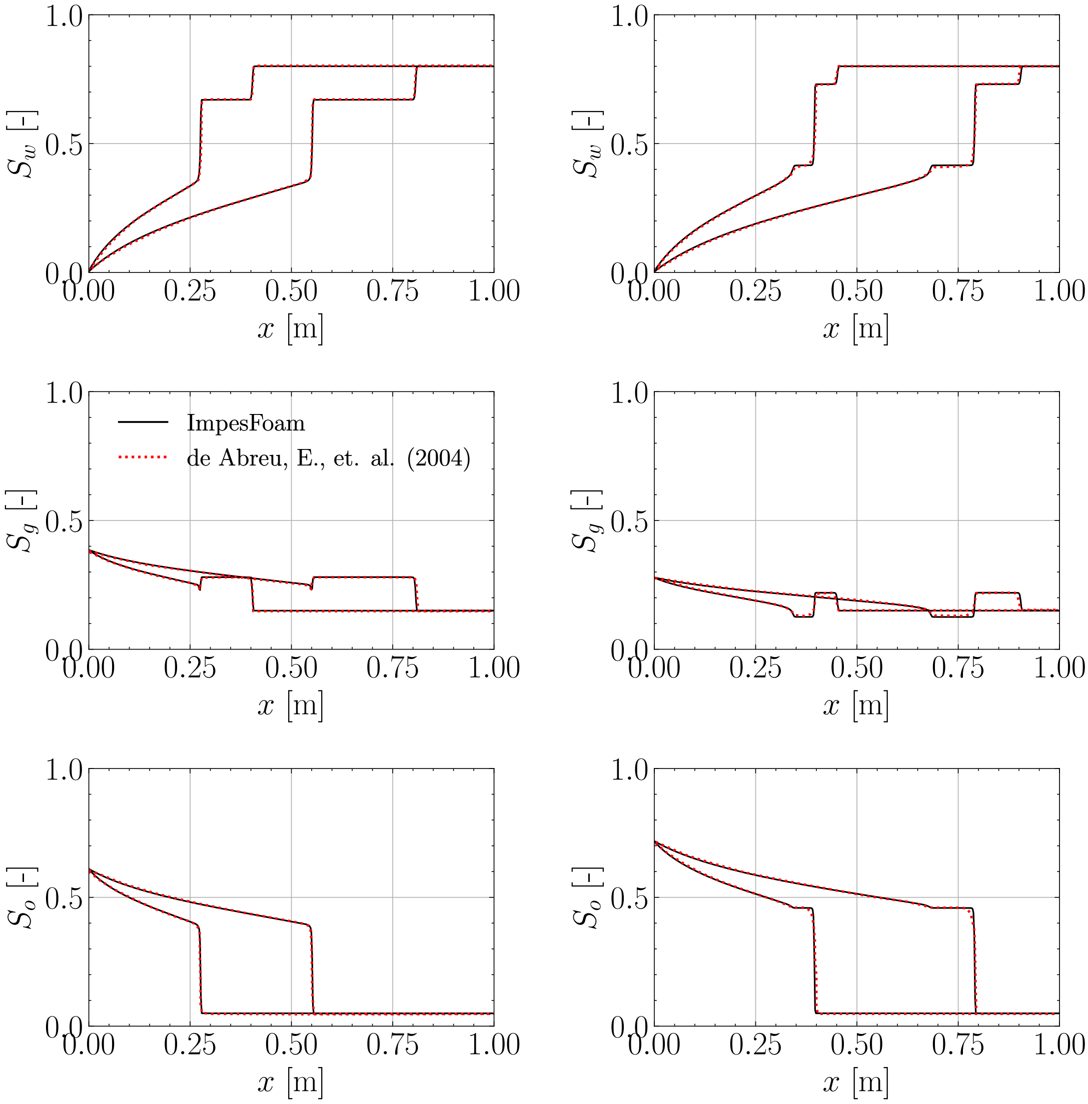


Figure 4: Comparison of one-dimensional displacement profiles obtained with ImpesFOAM against the reference solutions of Abreu et al. [2], for the case without gravity and capillary-pressure effects.

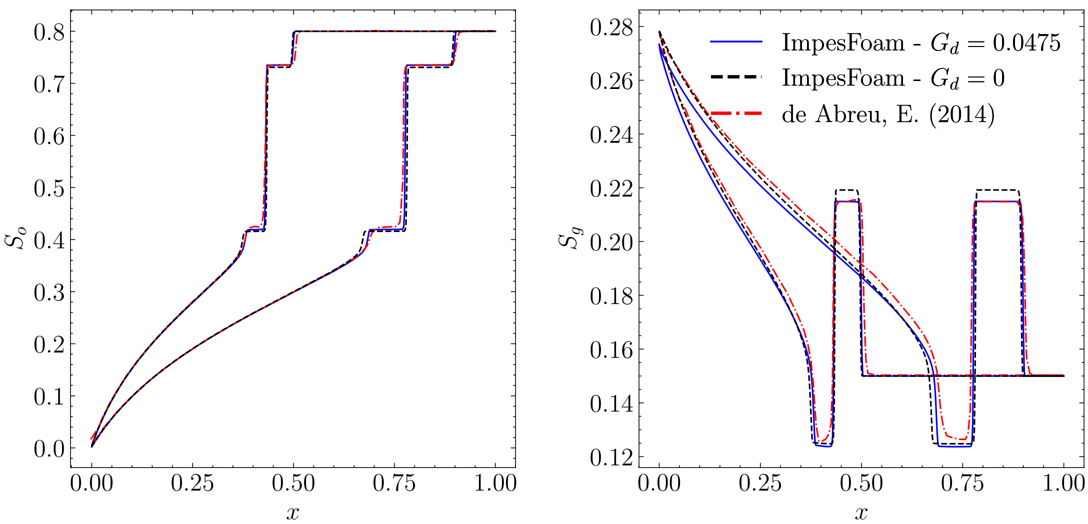


Figure 5: Comparison of solution profiles for one-dimensional displacement with $G_d = 0.0475$ between ImpesFOAM and results from Abreu [1].

the initial water saturation of the column is set far from its thermodynamic equilibrium in a step-wise fashion: the lower half is partially saturated with water ($S_w = 0.5$) while the upper half is initially dry as shown in Fig. 6. To ensure proper equilibriation, both fluids are allowed to flow freely through the column's top boundary, but not through its lower one, as depicted by the boundary conditions shown in Fig. 6. For this simplified case, the theoretical steady-state can be described as the balance between capillary and gravitational forces, where gravity pulls the heavier fluid (water) downwards while capillarity pulls it upwards, which is described by [14, 5]:

$$\frac{\partial p_c}{\partial x} = (\rho_g - \rho_w)\, g, \tag{S25}$$

which can be rearranged using the chain rule to yield:

$$\frac{\partial S_{w,pc}}{\partial x} = \frac{(\rho_g - \rho_w)\, g}{\frac{\partial p_c(S_{w,pc})}{\partial S_{w,pc}}}. \tag{S26}$$

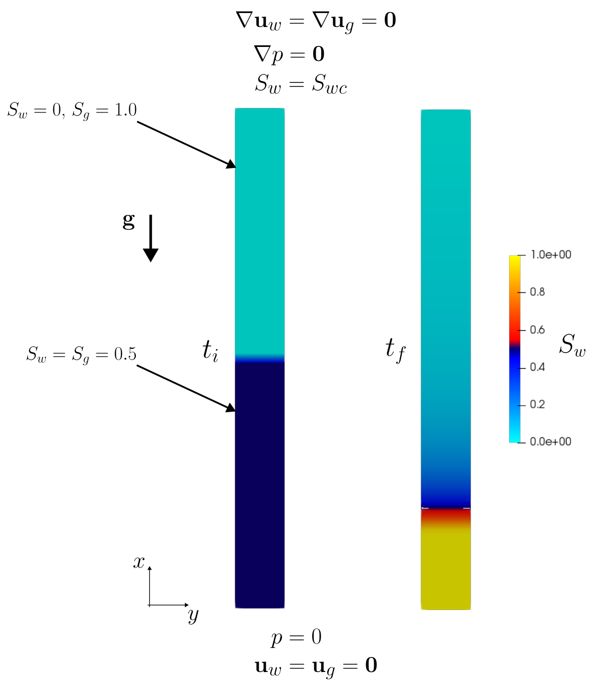


Figure 6: Schematic of the initial and boundary conditions used in the simulations. The figure shows the initial water saturation, S_w, distribution with which the porous column is saturated, t_i (left). The steady-state configuration of the domain is also illustrated at the final simulation time, t_f (left). The imposed lower and upper boundary conditions are indicated, as well as the direction of the gravity vector $\mathbf{g}$. Adapted from Carrillo et al. [5].

This steady-state saturation profile and $\frac{\partial S_{w,pc}}{\partial x}$ at steady-state are used as reference results. The capillary pressure correlations are based on the notion of effective saturation. However, the macro-scale capillary pressure tends to infinity when saturation S_w tends to S_{wc}. To accept irreducible and maximal saturations in numerical simulations, we define the effective saturation for capillary pressure $S_{w,pc}$ as follows [14]:

$$S_{w,pc} = \frac{S_w - S_{pc,irr}}{S_{pc,max} - S_{pc,irr}} \tag{S27}$$

where the minimal $S_{pc,irr}$ is a user-defined parameter that should satisfy $S_{pc,irr} < S_{wc}$. The correlation for capillary pressure proposed by Brooks and Corey [4] is given by:

$$p_c\,(S_{w,pc}) = p_{c,0} S_{w,pc}^{-\alpha} \tag{S28}$$

where $p_{c,0}$ is the entry capillary pressure and $1/\alpha$ the pore-size distribution index. Deriving Eq. (S28), $\frac{\partial p_c}{\partial S_b}$ can be expressed as:

$$\frac{\partial p_c}{\partial S_{w,pc}}\left(S_{w,pc}\right) = -\frac{\alpha p_{c,0}}{S_{pc,max} - S_{pc,irr}} S_{w,pc}^{-\alpha-1} \tag{S29}$$

The $p_c\left(S_{w,pc}\right)$ and $\frac{\partial p_c}{\partial S_{w,pc}}\left(S_{w,pc}\right)$ curves are show in Fig. 7 using parameters from Table 3, which were used in the numerical simulations.

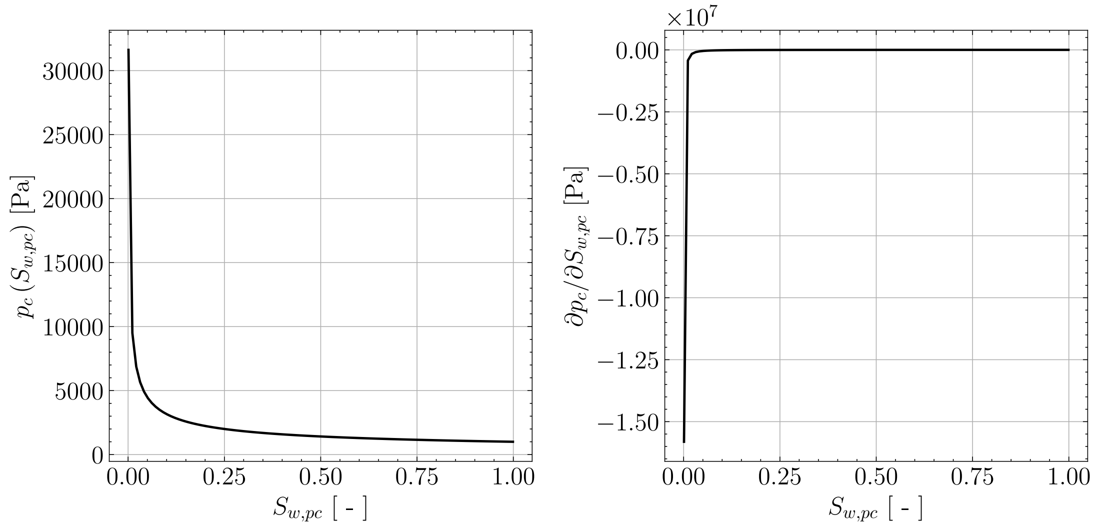


Figure 7: Capillary pressure $p_c\left(S_{w,pc}\right)$ curve (left) and its derivative $\frac{\partial p_c}{\partial S_{w,pc}}\left(S_{w,pc}\right)$ (right) using parameters from Table 3.

Parameter	**Value**	**Unit**	**Parameter**	**Value**	**Unit**
ρ_g	1.0	kg/m^3	ρ_w	1000	kg/m^3
μ_g	1.76×10^{-5}	Pa·s	μ_w	1.0×10^{-3}	Pa·s
g	9.81	m/s^2	ϕ	0.5	–
S_{wc}	1.0×10^{-3}	–	S_{gr}	1.0×10^{-3}	–
n_w	3.0	–	n_g	3.0	–
k_{rw}^0	1.0	–	k_{rg}^0	1.0	–
L	1.0	m	N_x	100	–
Δx	0.01	m	t_{final}	2.0×10^6	s
Δt	10.0	s			

Table 3: Simulation parameters for capillary pressure and gravity equilibrium benchmark.

Figure 8 compares the numerical results obtained with the Corey–Brooks relative permeability model against the analytical solution and other numerical results reported in the literature [5].

3.2. Three-phase flow with capillary pressure effects

Finally, we analyze RP_1 and RP_2 cases using the Leverett model for capillary pressure [16], while neglecting gravity effects. The Leverett curves are given by:

$$p_{cwo} = 5\epsilon(2 - S_w)(1 - S_w) \qquad \text{and} \qquad p_{cog} = 5\epsilon(2 - S_g)(1 - S_g), \tag{S30}$$

where ϵ represents the relative importance of advective and capillary/dispersive forces. It is defined as the inverse of Péclet number (Pe):

$$\epsilon = \frac{1}{Pe}, \qquad Pe = \frac{\nu_c \mu_o L}{p_c k_c}, \tag{S31}$$

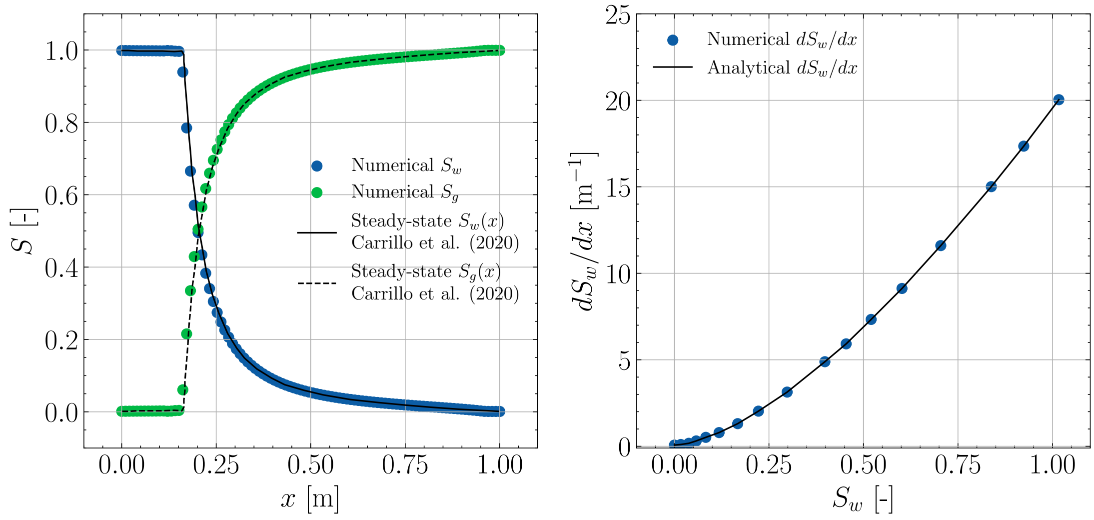


Figure 8: Comparison of steady-state water saturation profiles from our numerical (dots) framework with the analytical solution in Eq. (S26) and numerical results from literature [5]. The left panel shows the steady-state saturation profiles, while the right panel presents the corresponding equilibrium water saturation gradient for the Corey–Brooks capillary pressure model [4].

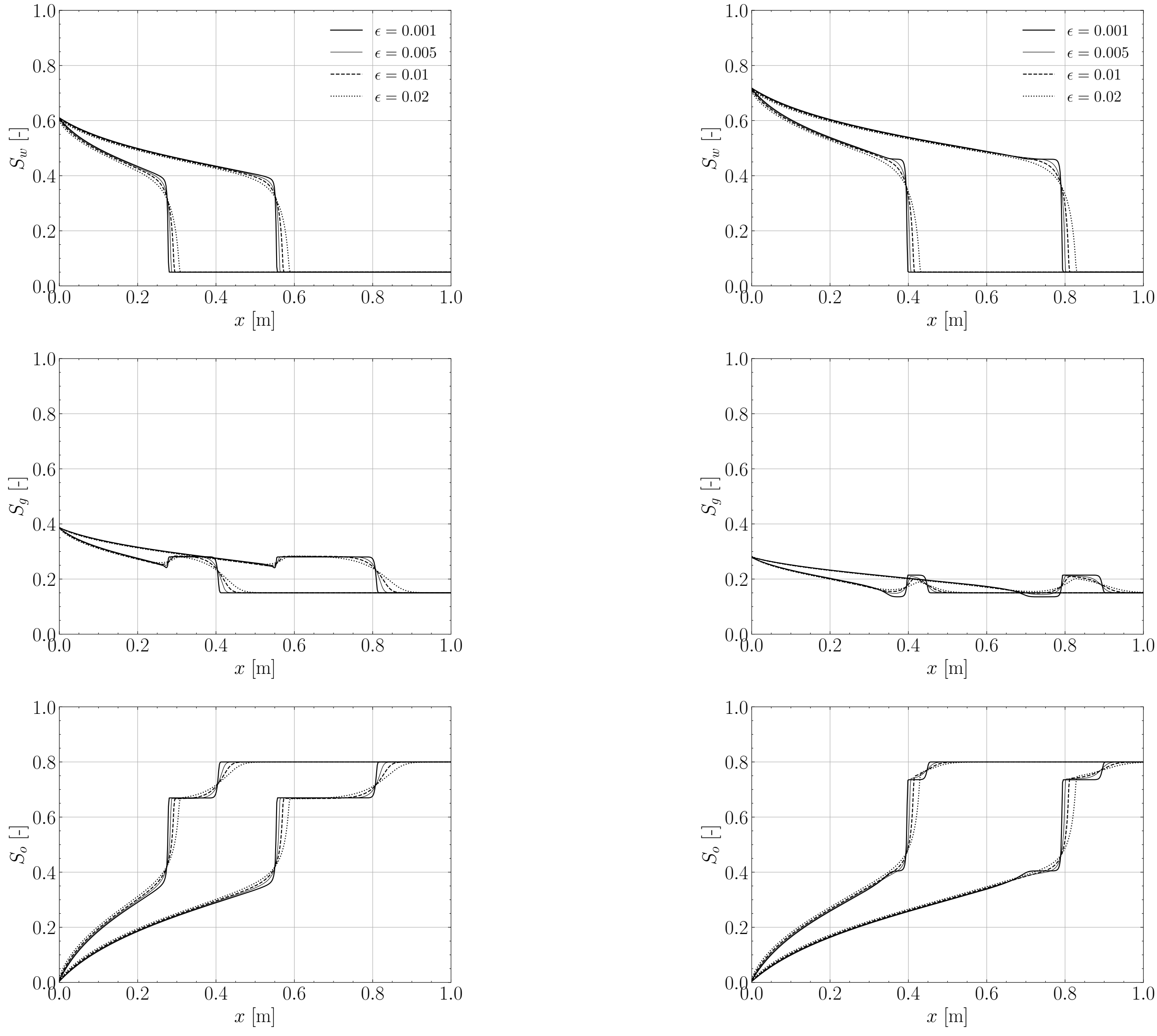
where k_c is the characteristic permeability, ν_c is the characteristic total velocity, μ_o the dynamic viscosity of the oil, L is the maximum characteristic length of the system, and p_c is the characteristic value for capillary pressures. Following the procedure adopted by Abreu et al. [2] to assess the influence of diffusive effects induced by capillary pressure, the parameter ϵ is varied over the values 0.001, 0.005, 0.01, and 0.02.

Figure 9 presents the solutions obtained on a grid with 500 cells. A clear qualitative resemblance can be observed between the solutions of RP_1 and RP_2 and those previously shown in Fig. 4. The main difference lies in the diffusive behavior, which becomes more pronounced as ϵ increases. This trend is consistent with the observations reported by Abreu et al. [2].

4. Benchmark case – Five-spot waterflood

Next we present a study case based on the laboratory 5-spot water flood described by Gaucher and Lindley [11], and hence also serves to validate the model with respect to source and sink as well as with scaled laboratory experiments. The 5-spot well pattern is depicted in Fig. 10. Water is injected into the injection wells (blue) to displace the resident oil towards the production wells (red). The parameters used for the simulations are shown in Table 5. Each well node was assigned only one-quarter of the given well fluxes due to symmetric considerations [18]. It is important to emphasize that the work of Panday et al. [18] only provides the values of relative permeabilities (k_{rw} and k_{ro}) data versus S_w data, as well as the residual saturations and endpoints. So it is necessary to fit the Corey-Brooks relative permeability exponents (n_w and n_o) using the reported data. The same holds for capillary pressure p_c versus S_w data. Figure 11 shows the data and the model with fitted parameters for relative permeability and capillary pressure functions.

In the scope of this work we do not focus on the wellbore modeling in porous media, considering only constant injection and extraction flow rates of wellbores. The reader is referred to the work of Peaceman [19], Chen et al. [6] and Quinelato et al. [20] for further details on this topic. It is important to note that the total flow rate q_t of all wells, both producers and injectors, is prescribed at all time steps. For injection wells, we have $q_t = q_w$ and $q_o = 0$, while for production wells, $q_t = q_w + q_o$. Thus, q_w is known at injection wells but not at production wells. In production wells, the water flow rate q_w is related to the total flow rate through a fractional flow function $f(S_w)$, such that: $q_w = f(S_w)q_t$, where $f(S_w)$ denotes the water fractional flow as



(a) RP_1

(b) RP_2

Figure 9: Solutions of the Riemann problems with capillary pressure effects for different values of ϵ.

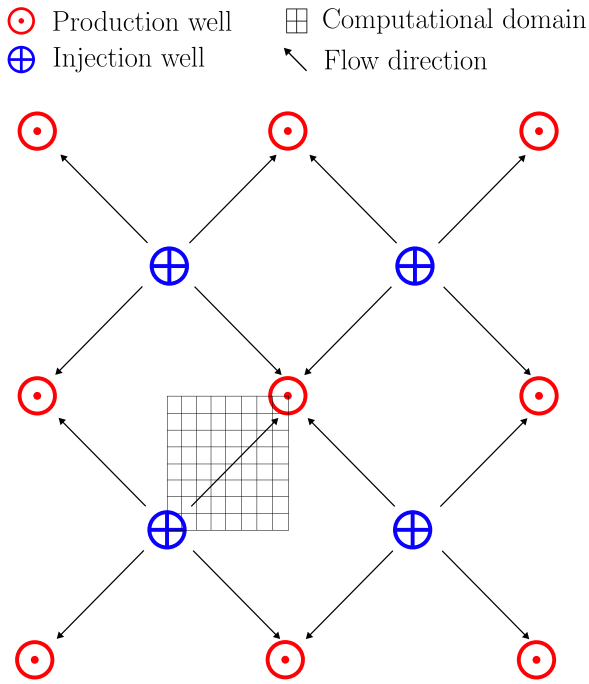


Figure 10: Schematic of the 5-spot water-flood problem. Adapted from [18, 11].

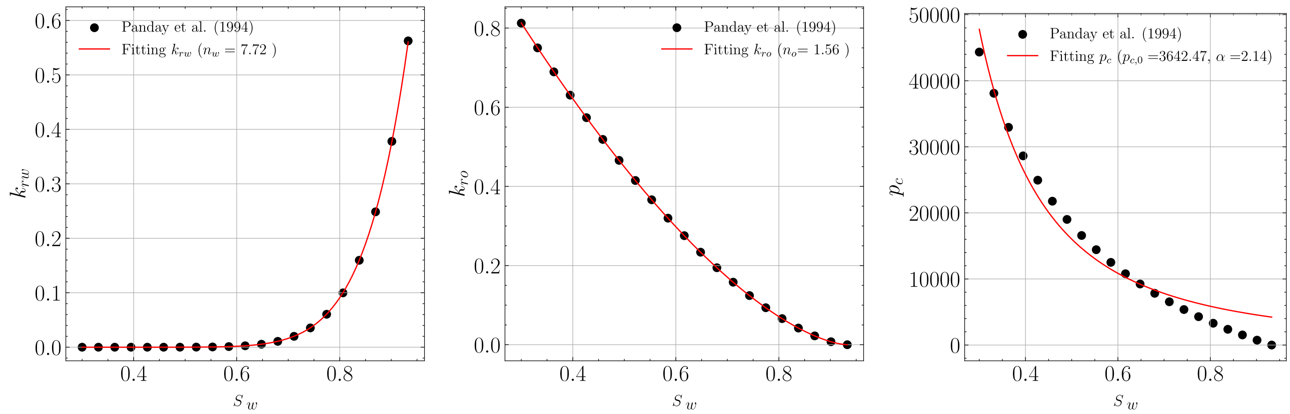


Figure 11: Fitting of n_w, n_o, $p_{c,0}$ and α from data [18].

a function of saturation S_w. In other words, q_w represents a fraction of the total flow rate q_t. Table 4 shows the values for q_w and q_o for both injection and production wells.

Wells	Water	Oil	Units
Injection	q_t	0	m^3 / $\mathrm{s} \cdot \mathrm{m}^3 \to 1/\mathrm{s}$
Production	$f_w \cdot qt$	$(1 - f_w) \cdot qt$	m^3 / $\mathrm{s} \cdot \mathrm{m}^3 \to 1/\mathrm{s}$

Table 4: Injection and production wells.

The simulation parameters are summarized in Table 5. Figure 12 presents the cumulative oil recovery, comparing experimental data from Gaucher and Lindley [11], numerical results from Panday et al. [18] obtained with finite-difference and finite-element methods, and the results of our simulations. The comparison indicates good agreement with literature. Finally, Fig. 13 illustrates the spatial distribution of water saturation at four time instants, highlighting the displacement dynamics within the domain.

Parameter	**Value**	**Unit**	**Parameter**	**Value**	**Unit**
ρ_o	800	$\mathrm{kg/m^3}$	ρ_w	1000	$\mathrm{kg/m^3}$
μ_o	2.17×10^{-3}	Pa·s	μ_w	0.5×10^{-3}	Pa·s
ϕ	0.2	–	S_{wc}	0.3	–
S_{or}	0.067	–	n_w	7.72 (fitted)	–
n_o	1.56 (fitted)	–	k_{rw}^0	0.562	–
k_{ro}^0	0.813	–	$p_{c,0}$	3642.47 (fitted)	Pa
α	2.14 (fitted)	–	S_{o_i}	0.7	–
q_t	3.1868×10^{-6}	1/s	t_{final}	2.0×10^6	s
$L_x = L_y$	142.245	m	$N_x = N_y$	50	–
$\Delta x = \Delta y$	≈ 2.85	m	t_{final}	5.0×10^8	s
Δt	1.0×10^4	s			

Table 5: Simulation parameters for the 5-spots benchmark.

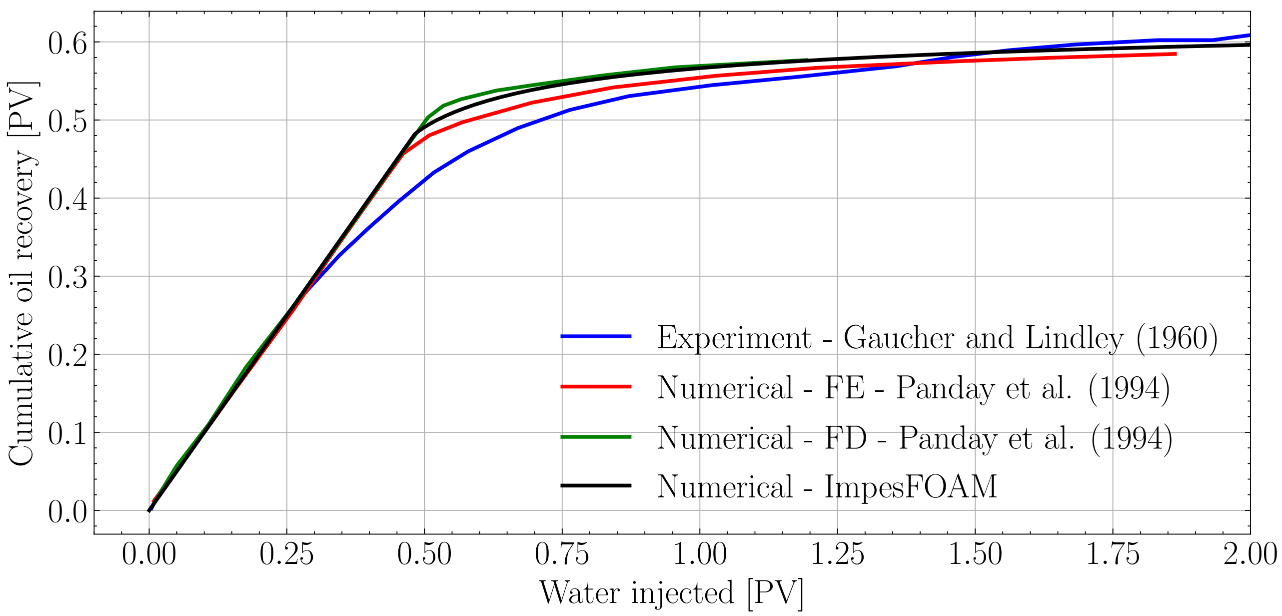


Figure 12: Comparison of cumulative oil recovery between the numerical results obtained in this work, those reported by Panday et al. [18], and the experimental data from Gaucher and Lindley [11].

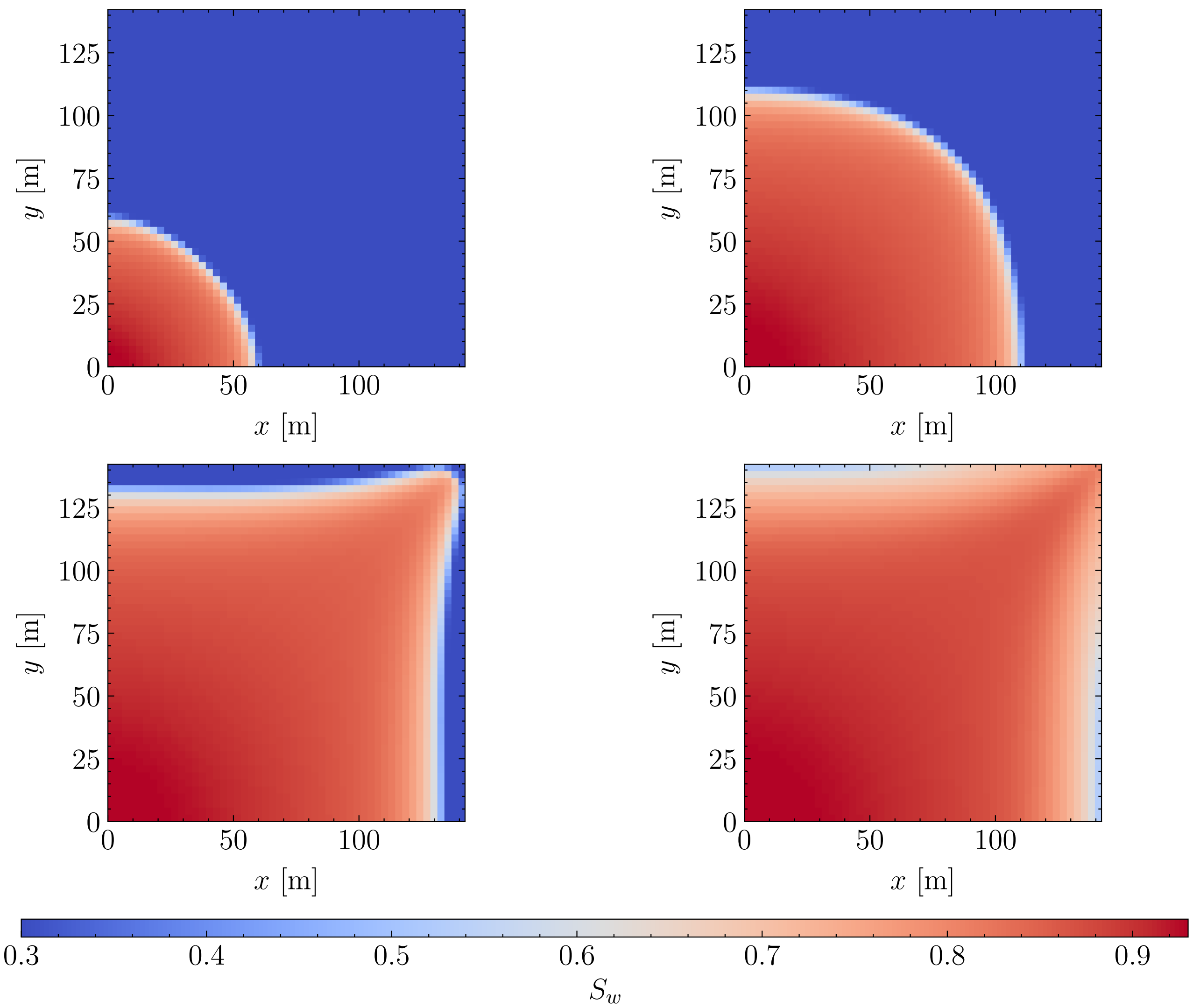


Figure 13: Spatial distribution of water saturation within the domain at four distinct time instants.

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